# Introduction to Progenetix and Cancer Classification

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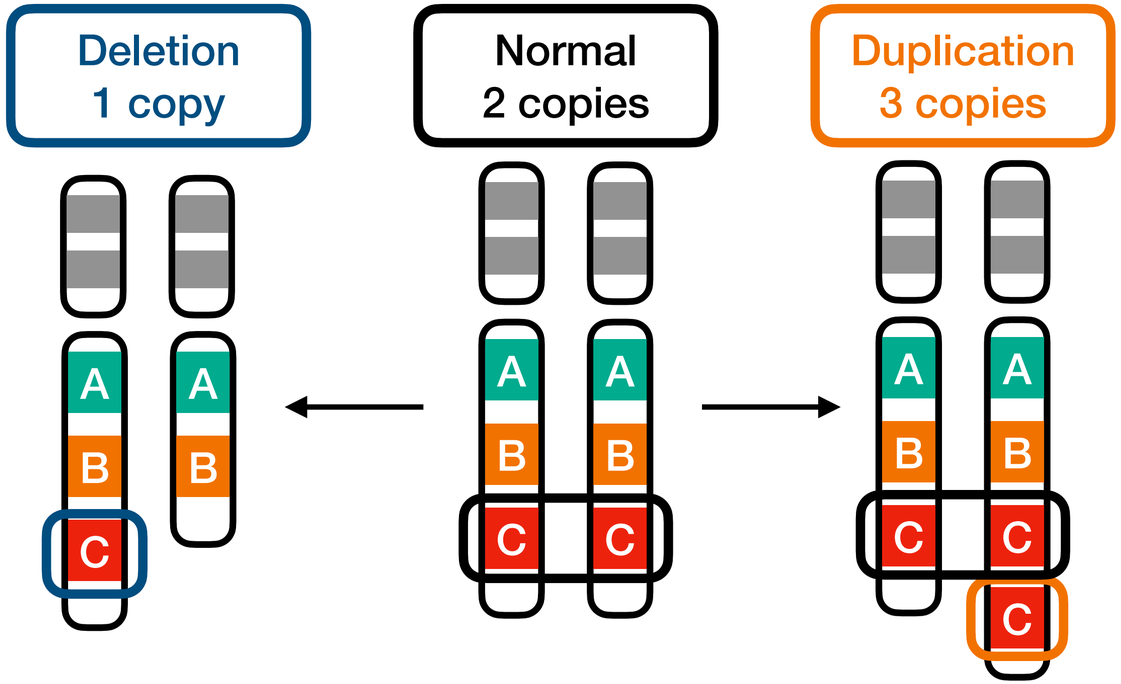
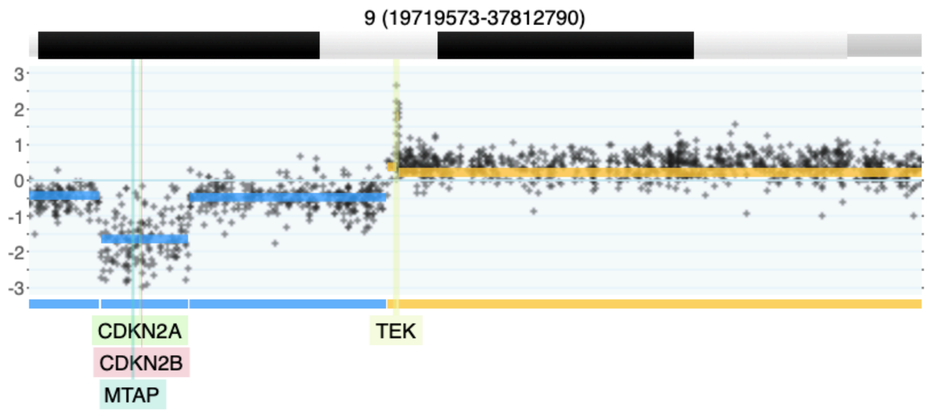
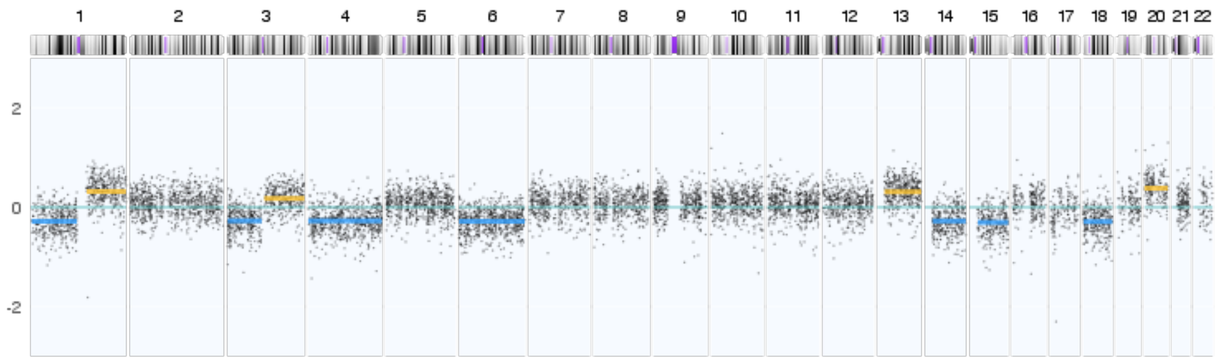
# Copy Number Variant (CNV)




- Intermediate-scale genetic change
- Size: 1kb to multiple megabase
- Additional copies of sequence (duplications) and losses of genetic material (deletions)




2-event, homozygous deletion in a Glioblastoma

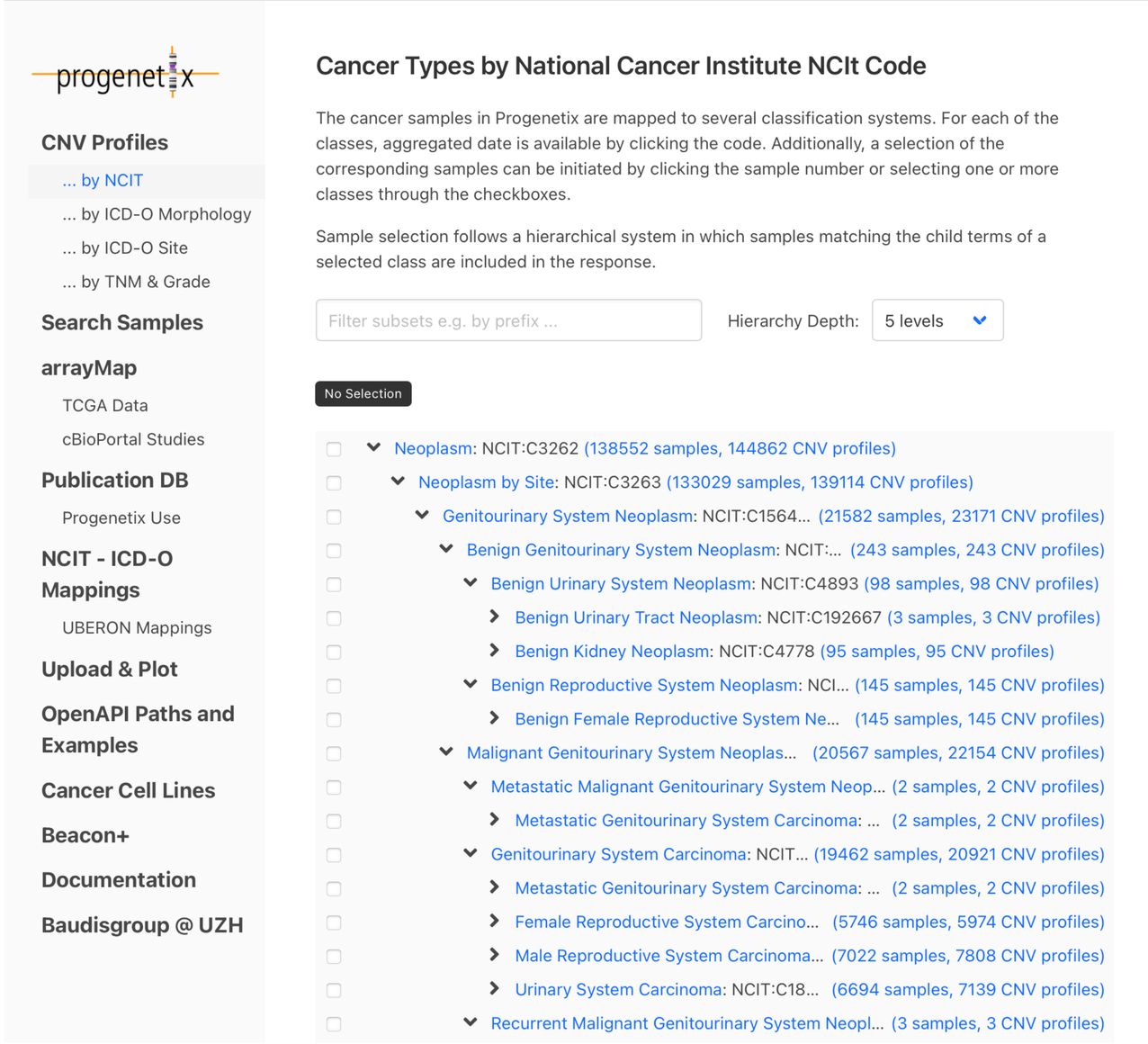



Gain of chromosome arm 13q in colorectal carcinoma

# progenetix.org

## Cancer Genomics Reference Resource

- ***open*** resource for oncogenomic profiles
- over **240'000 cancer CNV profiles**
- more than **1100 diagnostic types**
- runs on a **Beacon API**
- inclusion of reference datasets (e.g. TCGA)
- support for SNV data (TCGA, cell lines...)
- standardized encodings (e.g. NCIt, ICD-O 3)
- identifier mapping for PMID, GEO, Cellosaurus, TCGA, cBioPortal where appropriate
- core clinical data (TNM, sex, survival ...)
- data mapping services




Universität Zürich[UZH]

progenetix

SIB Swiss Institute of Bioinformatics

# How are cancer sample labels in Progenetix

- NCIt: hierarchical disease classification
- ICD-O Morphology: tumor type/morphology
- ICD-O Topography: anatomical site
- **NCIt** (NCI Thesaurus) combines disease concepts into one hierarchical label
- **ICD-O** (International Classification of Diseases for Oncology) separates morphology and anatomical site into two coding axes
- Example
  - pgx:icdom-81403: Adenocarcinoma, NOS
  - pgx:icdot-C56.9: Ovary
  - NCIT:C7700: Ovarian adenocarcinoma

# What kind of data do we get?

- **Sample data/metadata (Biosamples)**: one row per biosample
  - identifiers, diagnosis, sex, age, tissue, sample status, etc.

Results | Biosamples | Variants

| Biosample Id | Dx Classifications | Identifiers |
|---|---|---|
| pgxbs-m84dh8s2 | pgx:icdom-94403 Glioblastoma, NOS<br>pgx:icdot-C71.9 Brain, NOS<br>NCIT:C3058 Glioblastoma | |
| pgxbs-m84cz2cc | pgx:icdom-94403 Glioblastoma, NOS<br>pgx:icdot-C71.9 Brain, NOS<br>NCIT:C3058 Glioblastoma | |
| pgxbs-m84dd1zf | pgx:icdom-94403 Glioblastoma, NOS<br>pgx:icdot-C71.9 Brain, NOS<br>NCIT:C3058 Glioblastoma | |
| pgxbs-kftvil7b | pgx:icdom-94403 Glioblastoma, NOS<br>pgx:icdot-C71.9 Brain, NOS<br>NCIT:C3058 Glioblastoma | pubmed:18215105 Cloughesy TF, Yoshimoto K et al. (2008): Antitumor activity of rapamycin in a... |
| pgxbs-m84d05e1 | pgx:icdom-94403 Glioblastoma, NOS<br>pgx:icdot-C71.9 Brain, NOS<br>NCIT:C3058 Glioblastoma | |
| pgxbs-kftvku8q | pgx:icdom-94403 Glioblastoma, NOS<br>pgx:icdot-C71.9 Brain, NOS<br>NCIT:C3058 Glioblastoma | pubmed:28481359 Zehir A, Benayed R et al. (2017): Mutational landscape of metastatic cancer revealed...<br>cbioportal:msk_impact_2017 Targeted sequencing of 10,000 clinical cases using the MSK-IMPACT assay |
| pgxbs-m84d328c | pgx:icdom-94403 Glioblastoma, NOS<br>pgx:icdot-C71.9 Brain, NOS<br>NCIT:C3058 Glioblastoma | |
| pgxbs-kftvib0z | pgx:icdom-94403 Glioblastoma, NOS<br>pgx:icdot-C71.9 Brain, NOS<br>NCIT:C3058 Glioblastoma | pubmed:19074898 Solomon DA, Kim JS et al. (2008): Mutational inactivation of PTPRD in glioblastoma... |
| pgxbs-m84dguzb | pgx:icdom-94403 Glioblastoma, NOS<br>pgx:icdot-C71.9 Brain, NOS<br>NCIT:C3058 Glioblastoma | |
| pgxbs-kftvkuac | pgx:icdom-94403 Glioblastoma, NOS<br>pgx:icdot-C71.9 Brain, NOS<br>NCIT:C3058 Glioblastoma | pubmed:28481359 Zehir A, Benayed R et al. (2017): Mutational landscape of metastatic cancer revealed...<br>cbioportal:msk_impact_2017 Targeted sequencing of 10,000 clinical cases using the MSK-IMPACT assay |

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# What kind of data do we get?

- **Variants data:** one row per CNV segment
  - chromosome, start, end, CNV state

Results | Biosamples | Variants

| Digest | Gene | Pathogenicity | Variant type | Variant Instances |
|---|---|---|---|---|
| NC_000013.11:48957415-48970526:EFO_0020073 | | | CopyNumberChange | V: bycvar-6657d1aa02cc20f830464b3a<br>A: pgxcs-kftwy8s2<br>B: pgxbs-kftvkl70<br>I: pgxind-kftx6pk5 |
| NC_000001.11:77404639-77432003:EFO_0020073 | | | CopyNumberChange | V: bycvar-66582ac902cc20f8304c8969<br>A: pgxcs-kftx0dya<br>B: pgxbs-kftvkokk<br>I: pgxind-kftx6ts8 |
| NC_000005.10:102835209-108672054:EFO_0030068 | | | CopyNumberChange | V: bycvar-6657cb266ac278bc0e53cf82<br>A: pgxcs-kftwg76p<br>B: pgxbs-kftvj8w0<br>I: pgxind-kftx537c |
| NC_000004.12:182652615-190106768:EFO_0030068 | | | CopyNumberChange | V: bycvar-66e303cfd2ef81e071d26835<br>A: pgxcs-kftwtz7d<br>B: pgxbs-kftvksf6<br>I: pgxind-kftx6y7t |
| NC_000001.11:78243037-78282082:EFO_0030067 | | | CopyNumberChange | V: bycvar-5bab57a2727983b2e00d5552<br>A: pgxcs-kftvmnll<br>B: pgxbs-kftvgkdo<br>I: pgxind-kftx2asm |
| NC_000017.11:43934327-45395141:EFO_0030071 | | | CopyNumberChange | V: bycvar-665725b9183fc78d1fb12e88<br>A: pgxcs-kftw1tbd<br>B: pgxbs-kftvhxyr<br>I: pgxind-kftx3tkh |
| NC_000014.9:71698491-99196164:EFO_0030068 | | | CopyNumberChange | V: bycvar-665726b18f6b961582619a29<br>A: pgxcs-kftw3yi5<br>B: pgxbs-kftvi4nj<br>I: pgxind-kftx3vzk |
| NC_000008.11:11241951-11250222:EFO_0030070 | | | CopyNumberChange | V: bycvar-5bab57b6727983b2e00dfbc1<br>A: pgxcs-kftvmor0<br>B: pgxbs-kftvgkh1<br>I: pgxind-kftx2awp |
| NC_000023.11:41456779-41579686:EFO_0030071 | | | CopyNumberChange | V: bycvar-6656ffcbea1558707a981155<br>A: pgxcs-kftvz2p3<br>B: pgxbs-kftvhp46<br>I: pgxind-kftx3hdk |
| NC_000017.11:45382635-46062635:EFO_0030071 | | | CopyNumberChange | V: bycvar-673cf6922cbf8b196b9d26a7<br>A: pgxcs-m3io6eby<br>B: pgxbs-m3io6eby<br>I: pgxind-m3io2hwq |

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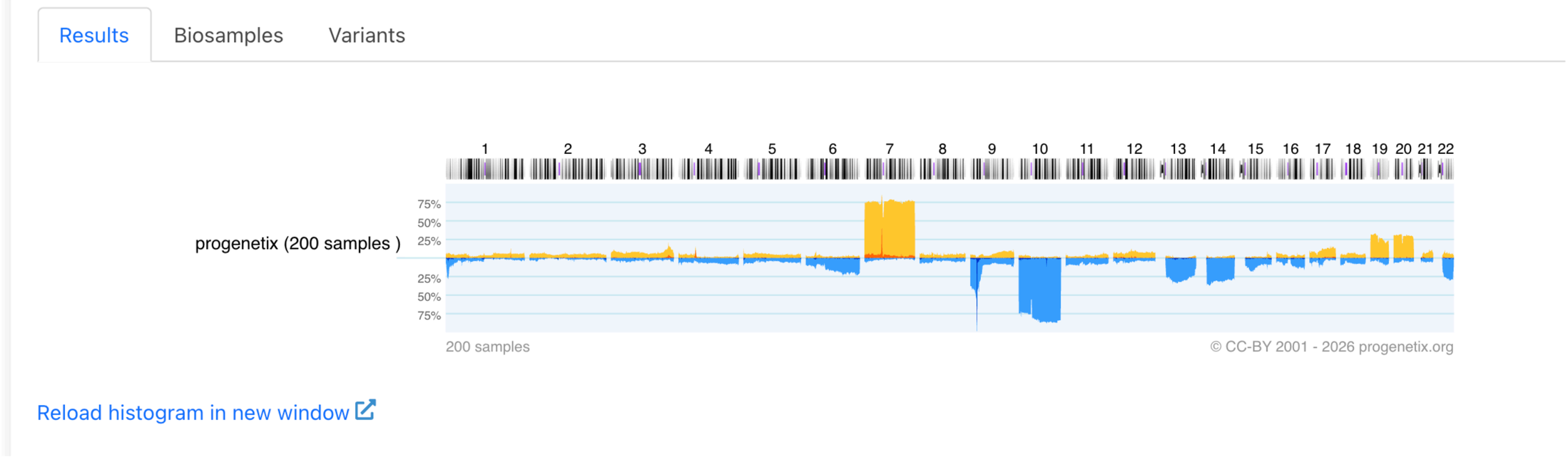
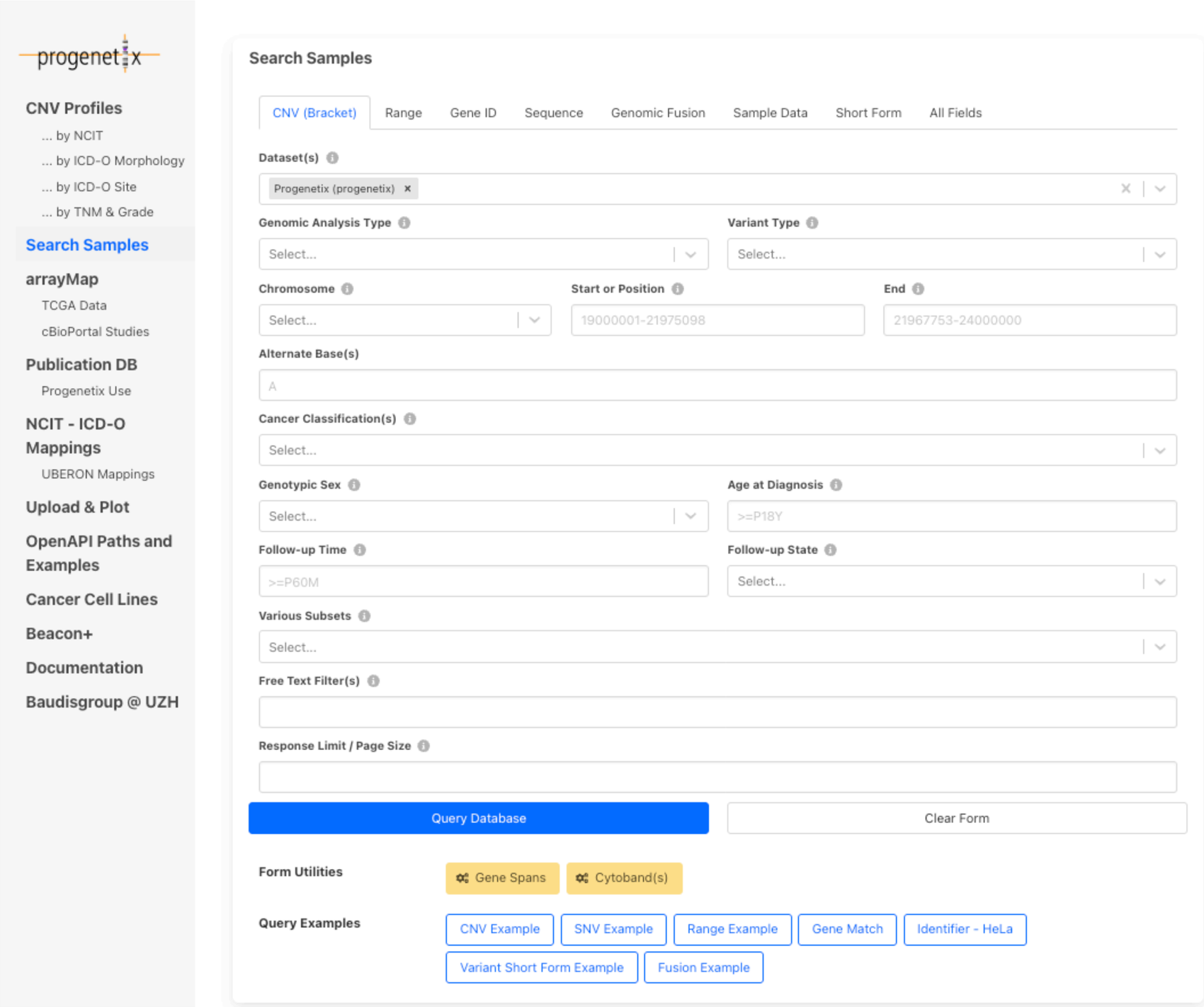
# What kind of data do we get?

- **Frequency data**: group-level CNV summaries across many samples
  - default **1 Mb binned** representation



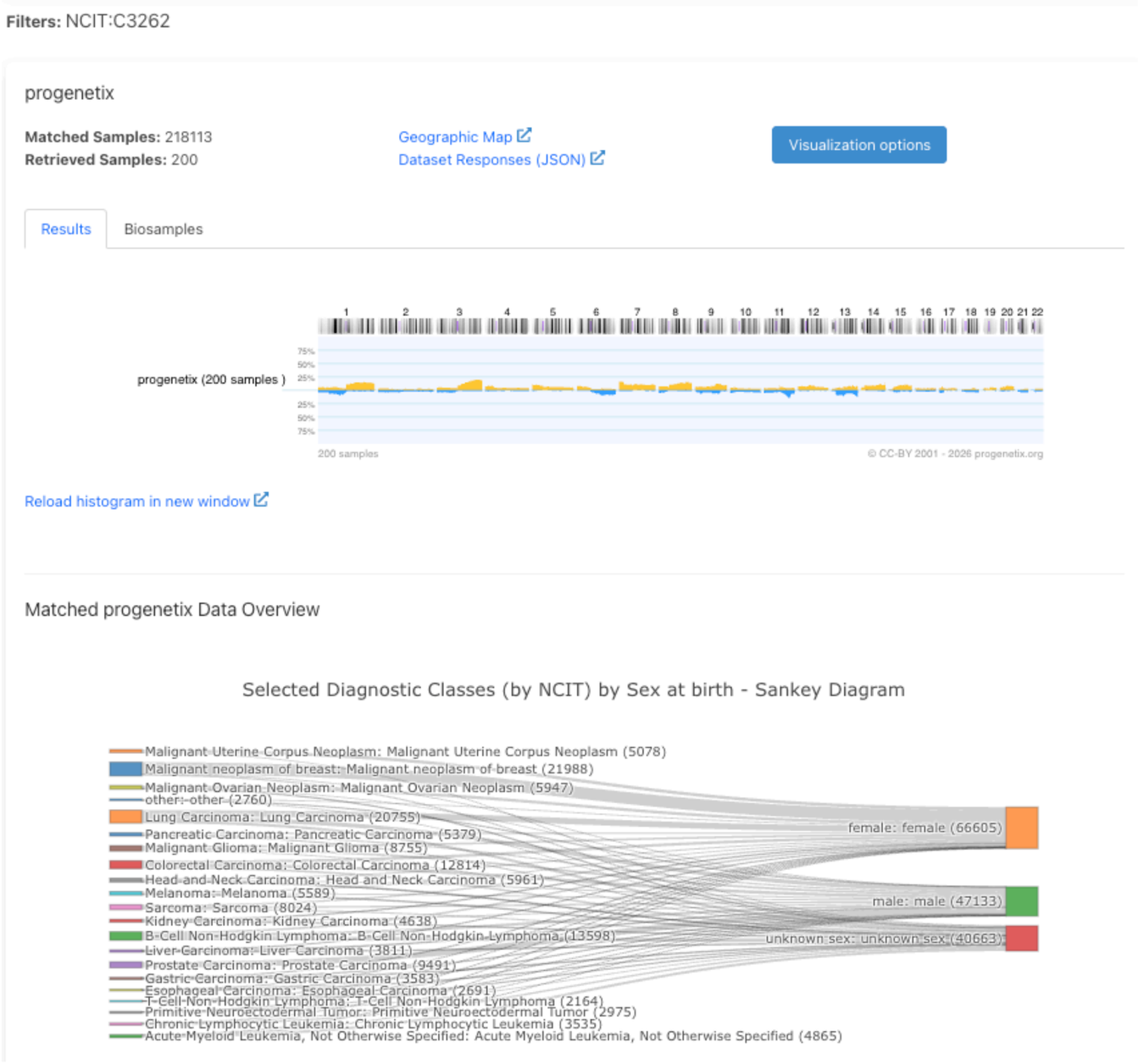
# Explore Progenetix Webpage

progenetix.org



# Small Task

- **Task 1**
- Find One cancer type of your choice in Progenetix
  - Write down:
    - Its classification label
    - The number of matched samples or profiles
    - One broader parent category, if shown
- **Task2**
- Search for one gene, for example, CDKN2A
- Check what kind of CNV-related query result is shown



# Explore Progenetix Dataset

- Download and Go through the notebook below:
- https://github.com/compbiozurich/UZH-BIO392-FS26/blob/master/course-material/2026-04-16_day3_exercise/intro_progenetix.ipynb
- Finish exercises.

# Read this paper and think about the following questions:

1. Why is important to construct a CNV map on health individuals of various ethnicities? **(Introduction)**
2. What is the CNV size that the authors defined? (Box 1 mentioned in **introduction**)

**Noted:** The CNV size definition is still under debate and may be different in other literatures

3. What are the primary approaches used for CNV detection? And what are the advantages and limitations of these technologies? **(CNV discoveries)**
4. The authors used clustering method to combine data from different studies into merged CNVRs (Copy number variable regions). What are the two criteria for cluster filtering? And why did they do this filtering? **(The CNV map)**
5. What are thresholds in stringency level 1, inclusive map (stringency level 2), and stringent map (stringency level 12) respectively? **(The CNV map)**
6. Which percentage of the genome contributes to CNV in inclusive map and stringent map respectively? (**Properties of the CNV map**)
7. By your intuition, which kind of genes are more variable between protein-coding genes and non-coding genes? How about their findings in this paper? **(Functional impact of CNV)**
8. The authors generated a null CNV map and found genes for which at least 85% of the exons were homozygous deleted. What are the functions of these genes? And why did the authors say that they seem to be non-essential? **(Homozygous deleted genes)**
9. If you are a medical doctor, how do you use this map as a tool to assess the clinical importance of a CNV? **(Clinical application of the CNV map part in Discussion)**

nature > nature reviews genetics > analyses > article

Published: 03 February 2015

## A copy number variation map of the human genome

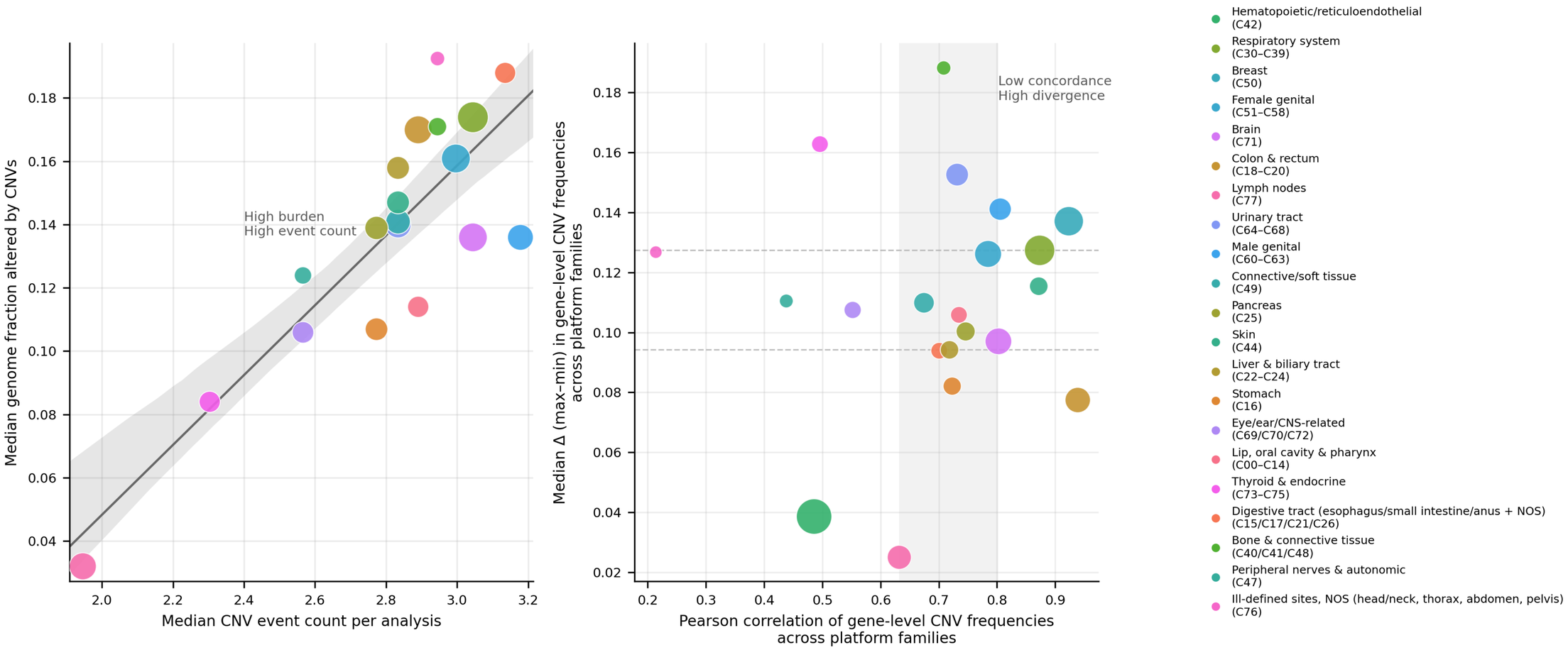
Mehdi Zarrei, Jeffrey R. MacDonald, Daniele Merico & Stephen W. Scherer

*Nature Reviews Genetics* **16**, 172–183 (2015) | Cite this article

**29k** Accesses | **400** Citations | **63** Altmetric | Metrics

Paper link: https://www.nature.com/articles/nrg3871

# Why cancer classfication



# Why cancer classfication

- Group-level comparison tells us that cancer CNV patterns are different.
- The next question is whether these differences can classify individual samples.
- Existing cancer labels are useful, but they still need to be examined in genomic data.
- Classification helps us connect biological heterogeneity, feature design, and interpretation.

# Classfication Showcase

- Download and Go through the notebook below:
- https://github.com/compbiozurich/UZH-BIO392-FS26/blob/master/course-material/2026-04-16_day3_exercise/intro_classification.ipynb
- Finish exercises.